Forward propagation and backpropagation

This exercise asks for the forward and backward pass of a small multilayer perceptron, first for one training pair and then in matrix form for a minibatch. The calculus is the chain rule; the linear algebra is there to keep the shapes honest. A calculator is appropriate for the logistic function. Every derivative should be derived, not quoted from a library.

1 Setup

Consider a network with input dimension 2, one hidden layer of width 2, and a scalar output. Both layers use the logistic sigmoid

$$\sigma(t) = \frac{1}{1+e^{-t}}, \qquad \sigma'(t) = \sigma(t)\big(1-\sigma(t)\big),$$

applied elementwise to vectors. For an input $x \in \mathbb{R}^2$ and a target $y \in \mathbb{R}$,

$$z^{(1)} = W^{(1)}x + b^{(1)}, \qquad a^{(1)} = \sigma(z^{(1)}), \tag{1}$$

$$z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}, \qquad \hat{y} = \sigma(z^{(2)}), \tag{2}$$

$$L = \tfrac{1}{2}(\hat{y} - y)^2. \tag{3}$$

The parameters of the worked example are

$$W^{(1)} = \begin{pmatrix} 0.5 & -0.2 \\ 0.1 & 0.3 \end{pmatrix}, \qquad b^{(1)} = \begin{pmatrix} 0.1 \\ -0.1 \end{pmatrix},$$

$$W^{(2)} = \begin{pmatrix} 0.4 & -0.5 \end{pmatrix}, \qquad b^{(2)} = 0.2,$$

and the training pair is $x = (1,\, 0)^{\mathsf{T}}$, $y = 1$. Hidden units are written h_1, h_2 in the diagram; their values are the coordinates of $a^{(1)}$.

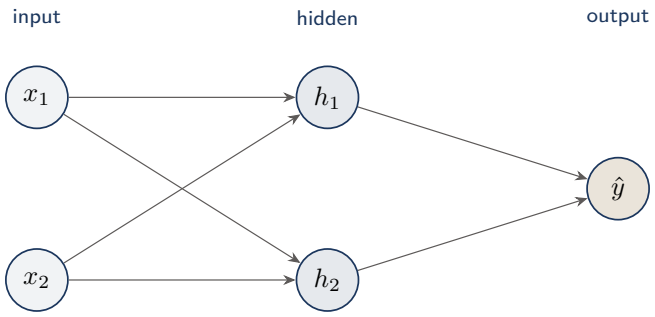


Figure 1. The 2–2–1 network. Biases are omitted from the drawing and appear in (1)–(2).

Throughout, vectors are columns. The gradient of a scalar with respect to a matrix W is the matrix of partial derivatives, of the same shape as W. The symbol $\odot$ denotes the elementwise (Hadamard) product.

2 Questions

(a) Write the forward computation in the shapes of (1)–(3) and evaluate $z^{(1)}$, $a^{(1)}$, $z^{(2)}$, $\hat{y}$, and L to six decimal places.

(b) Let $\delta^{(2)} = \partial L/\partial z^{(2)}$, a scalar. Derive

$$\frac{\partial L}{\partial W^{(2)}} = \delta^{(2)}\big(a^{(1)}\big)^{\mathsf{T}}, \qquad \frac{\partial L}{\partial b^{(2)}} = \delta^{(2)}$$

from the chain rule, and evaluate both quantities.

(c) Show that the Jacobian of $u \mapsto \sigma(u)$ at $z^{(1)}$ is diagonal. Deduce

$$\delta^{(1)} = \big(W^{(2)}\big)^{\mathsf{T}}\delta^{(2)} \odot \sigma'\big(z^{(1)}\big) \in \mathbb{R}^2$$

and the hidden-layer gradients

$$\frac{\partial L}{\partial W^{(1)}} = \delta^{(1)}x^{\mathsf{T}}, \qquad \frac{\partial L}{\partial b^{(1)}} = \delta^{(1)}.$$

Evaluate them, and explain the zero pattern in $\partial L/\partial W^{(1)}$ without referring to the numerical values of the sigmoids.

(d) Now let a minibatch consist of N examples stored as columns of $X \in \mathbb{R}^{d_0 \times N}$, and let the training objective be the average of the per-example losses. For each layer ℓ, with preactivations $Z^{(\ell)}$ and activations $A^{(\ell)}$ likewise stored by columns, prove

$$\frac{\partial L}{\partial W^{(\ell)}} = \frac{1}{N}\Delta^{(\ell)}\big(A^{(\ell-1)}\big)^{\mathsf{T}}, \qquad \frac{\partial L}{\partial b^{(\ell)}} = \frac{1}{N}\Delta^{(\ell)}\mathbf{1}, \tag{4}$$

where column n of $\Delta^{(\ell)}$ is the gradient of the *per-example* loss with respect to $z_n^{(\ell)}$, and $\mathbf{1} \in \mathbb{R}^N$ is the all-ones vector. Then specialise the output factor: for the squared loss (3) it retains a factor $\sigma'(z)$, whereas for the binary cross-entropy

$$\ell(\hat{y}, y) = -\,y \log \hat{y} - (1-y)\log(1-\hat{y})$$

with $\hat{y} = \sigma(z)$ and $y \in \{0, 1\}$, the output delta collapses to $\hat{y} - y$.

(e) Let θ collect every parameter and let $g = \nabla_\theta L(\theta)$. Define $\varphi(\eta) = L(\theta - \eta g)$. Show that $\varphi'(0) = -\|g\|_2^2$, and conclude that if $g \neq 0$ then every sufficiently small positive step strictly decreases L. For the numerical network, take $\eta = 1/2$, report the updated parameters to six decimal places, recompute the loss, and check the decrease against the first-order prediction $\eta\|g\|_2^2$.

The worked solution begins on the next page.

3 Solutions

Numerical entries below are correctly rounded to six decimal places from double-precision evaluation. Every nonzero partial derivative agrees with a central difference of step 10^{-6} to a relative error below 10^{-9}. The structural zeros are exact.

Solution to (a)

The shapes are those in (1)–(3): $W^{(1)} \in \mathbb{R}^{2\times 2}$, $b^{(1)} \in \mathbb{R}^2$, $W^{(2)} \in \mathbb{R}^{1\times 2}$, and $b^{(2)} \in \mathbb{R}$. Substituting the given sample,

$$z^{(1)} = \begin{pmatrix} 0.5 & -0.2 \\ 0.1 & 0.3 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0.1 \\ -0.1 \end{pmatrix} = \begin{pmatrix} 0.6 \\ 0 \end{pmatrix}.$$

The second coordinate vanishes because it is the second standard basis direction: only the first column of $W^{(1)}$ meets a nonzero input. Then

$$a^{(1)} = \begin{pmatrix} \sigma(0.6) \\ \sigma(0) \end{pmatrix} = \begin{pmatrix} 0.645656 \\ 0.500000 \end{pmatrix},$$

since $\sigma(0) = 1/2$ exactly and $\sigma(0.6) = 0.645656306\ldots$. The output preactivation is

$$\begin{aligned} z^{(2)} &= 0.4 \cdot 0.645656306 - 0.5 \cdot 0.5 + 0.2 \\ &= 0.208262522, \end{aligned}$$

so $z^{(2)} = 0.208263$ to six decimals, and

$$\hat{y} = \sigma\big(z^{(2)}\big) = 0.551878, \qquad L = \tfrac{1}{2}\big(0.551878255 - 1\big)^2 = 0.100407.$$

The prediction sits well below the target, which will determine the sign of every output-layer derivative.

Solution to (b)

Differentiate the scalar map $z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$ with $a^{(1)}$ held fixed. If $W^{(2)} = (w_1,\, w_2)$ and $a^{(1)} = (a_1,\, a_2)^{\mathsf{T}}$, then $z^{(2)} = w_1a_1 + w_2a_2 + b^{(2)}$, so

$$\frac{\partial z^{(2)}}{\partial w_j} = a_j, \qquad \frac{\partial z^{(2)}}{\partial b^{(2)}} = 1.$$

The chain rule through L contributes the same upstream factor $\delta^{(2)} = \partial L/\partial z^{(2)}$ to each parameter, and the two partials in w assemble into the outer product $\delta^{(2)}(a^{(1)})^{\mathsf{T}}$. In differential notation, $\mathrm{d}z^{(2)} = \mathrm{d}W^{(2)}\, a^{(1)} + \mathrm{d}b^{(2)}$ and

$$\mathrm{d}L = \delta^{(2)}\, \mathrm{d}z^{(2)} = \big\langle \delta^{(2)}(a^{(1)})^{\mathsf{T}},\, \mathrm{d}W^{(2)} \big\rangle_F + \delta^{(2)}\, \mathrm{d}b^{(2)},$$

which is the same identity. For the squared loss,

$$\frac{\partial L}{\partial \hat{y}} = \hat{y} - y, \qquad \delta^{(2)} = (\hat{y} - y)\, \sigma'\big(z^{(2)}\big).$$

Numerically $\hat{y} - y = -0.448122$ and

$$\sigma'\big(z^{(2)}\big) = \hat{y}(1 - \hat{y}) = 0.247309,$$

hence $\delta^{(2)} = -0.110824$. The minus sign says that increasing $z^{(2)}$ decreases L, as it must while $\hat{y} < y$. Therefore

$$\begin{aligned} \frac{\partial L}{\partial W^{(2)}} &= -0.110824 \cdot \begin{pmatrix} 0.645656 & 0.500000 \end{pmatrix} = \begin{pmatrix} -0.071554 & -0.055412 \end{pmatrix}, \\ \frac{\partial L}{\partial b^{(2)}} &= -0.110824. \end{aligned}$$

Solution to (c)

Write $\sigma(u)_i = \sigma(u_i)$. Then

$$\frac{\partial \sigma(u)_i}{\partial u_j} = \sigma'(u_i)\,\mathbf{1}_{\{i=j\}},$$

so the Jacobian $D\sigma(z^{(1)})$ is the diagonal matrix $\operatorname{diag}(\sigma'(z^{(1)}))$. No hidden unit's output depends on another unit's preactivation.

The output preactivation depends on the hidden activation through $z^{(2)} = W^{(2)}a^{(1)} + b^{(2)}$. With $W^{(2)}$ a row, the gradient of $z^{(2)}$ with respect to the column $a^{(1)}$ is $(W^{(2)})^\mathsf{T}$. The chain rule from $z^{(2)}$ down to $z^{(1)}$ is therefore

$$\delta^{(1)} = \frac{\partial L}{\partial z^{(1)}} = D\sigma\big(z^{(1)}\big)^\mathsf{T}\big(W^{(2)}\big)^\mathsf{T}\delta^{(2)}. \tag{5}$$

The Jacobian is symmetric, and multiplication by a diagonal matrix is the Hadamard product with its diagonal, which is (5) in the form requested. Explicitly,

$$\big(W^{(2)}\big)^\mathsf{T}\delta^{(2)} = \begin{pmatrix} 0.4 \\ -0.5 \end{pmatrix}(-0.110824) = \begin{pmatrix} -0.044330 \\ 0.055412 \end{pmatrix}.$$

The hidden derivatives are $\sigma'(0.6) = 0.228784$ and $\sigma'(0) = 1/4$, so

$$\delta^{(1)} = \begin{pmatrix} -0.044330 \cdot 0.228784 \\ 0.055412 \cdot 0.25 \end{pmatrix} = \begin{pmatrix} -0.010142 \\ 0.013853 \end{pmatrix}.$$

The same outer-product argument as in (b), now with incoming activation x, gives

$$\frac{\partial L}{\partial W^{(1)}} = \delta^{(1)}x^\mathsf{T} = \begin{pmatrix} -0.010142 \\ 0.013853 \end{pmatrix}\begin{pmatrix} 1 & 0 \end{pmatrix} = \begin{pmatrix} -0.010142 & 0 \\ 0.013853 & 0 \end{pmatrix},$$

and $\partial L/\partial b^{(1)} = \delta^{(1)}$.

The second column is zero for a structural reason. The forward map uses $W^{(1)}$ only through the product $W^{(1)}x$. Differentiating in an entry $W^{(1)}_{ij}$ produces a multiple of the coordinate x_j. Here $x_2 = 0$, so every weight that multiplies x_2 can be changed without changing $z^{(1)}$, $a^{(1)}$, $\hat{y}$, or L. Those partial derivatives vanish before any activation is evaluated. A nonzero input coordinate would not have produced this column of zeros; saturation of σ is a different phenomenon, visible here only in the magnitudes of σ'.

Solution to (d)

Minibatch outer products. Let ℓ_n be the loss of example n, and let $L = N^{-1}\sum_{n=1}^{N}\ell_n$. Write $A^{(0)} = X$ and

$$Z^{(\ell)} = W^{(\ell)}A^{(\ell-1)} + b^{(\ell)}\mathbf{1}^\mathsf{T}, \qquad A^{(\ell)} = \sigma\big(Z^{(\ell)}\big),$$

with σ applied elementwise. Let $S = \sum_n \ell_n$, and let column n of $\Delta^{(\ell)}$ be $\partial\ell_n/\partial z_n^{(\ell)}$. The scalar S depends on the preactivation matrix through

$$\mathrm{d}S = \big\langle \Delta^{(\ell)},\, \mathrm{d}Z^{(\ell)} \big\rangle_F$$

when the differential is taken at layer ℓ, because each column contributes its own inner product and distinct examples do not share preactivations. Holding the incoming activation fixed,

$$\mathrm{d}Z^{(\ell)} = \mathrm{d}W^{(\ell)}\,A^{(\ell-1)} + \mathrm{d}b^{(\ell)}\,\mathbf{1}^\mathsf{T}.$$

The Frobenius identity $\langle \Delta,\, \mathrm{d}W\,A\rangle_F = \langle \Delta A^\mathsf{T},\, \mathrm{d}W\rangle_F$ then yields

$$\frac{\partial S}{\partial W^{(\ell)}} = \Delta^{(\ell)}\big(A^{(\ell-1)}\big)^\mathsf{T}, \qquad \frac{\partial S}{\partial b^{(\ell)}} = \Delta^{(\ell)}\mathbf{1}.$$

Dividing by N is (4). The recurrence that builds $\Delta^{(\ell)}$ from $\Delta^{(\ell+1)}$ is the columnwise form of (5),

$$\Delta^{(\ell)} = \left((W^{(\ell+1)})^{\mathsf{T}}\Delta^{(\ell+1)}\right) \odot \sigma'(Z^{(\ell)}),$$

with the output case supplied by the choice of ℓ_n. The factor $1/N$ is not part of this recurrence: it appears once, when the averaged objective differentiates with respect to a parameter.

One orientation warning is worth recording. The convention here stores one example per column, so the activation matrix is features-by-batch. Implementations that store one example per row use the transposed product, with the factor $1/N$ unchanged.

Squared loss and cross-entropy. For $\ell = \frac{1}{2}(\hat{y} - y)^2$ and $\hat{y} = \sigma(z)$,

$$\frac{\partial \ell}{\partial z} = (\hat{y} - y)\,\sigma'(z),$$

which is the output column of $\Delta^{(L)}$. For binary cross-entropy, differentiate in $\hat{y}$ first:

$$\frac{\partial \ell}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{1-\hat{y}} = \frac{\hat{y} - y}{\hat{y}\,(1-\hat{y})},$$

provided $\hat{y} \in (0,1)$. Multiply by $\sigma'(z) = \hat{y}(1-\hat{y})$:

$$\frac{\partial \ell}{\partial z} = \frac{\hat{y} - y}{\hat{y}\,(1-\hat{y})} \cdot \hat{y}\,(1-\hat{y}) = \hat{y} - y. \tag{6}$$

The sigmoid derivative cancels, including at the saturated tails where $\sigma'(z)$ itself underflows. The same algebra holds for a soft target $y \in (0,1)$; the hypothesis $y \in \{0,1\}$ is only the classification interpretation. Combining (6) with (4), a sigmoid output trained by mean binary cross-entropy has

$$\frac{\partial L}{\partial W^{(L)}} = \frac{1}{N}(\hat{y} - y)\,\left(A^{(L-1)}\right)^{\mathsf{T}}$$

in the scalar-output case, with $\hat{y} - y$ the row of residuals. The squared loss keeps an extra diagonal factor $\sigma'(z^{(L)})$ and is therefore more prone to stalling once the output saturates. That comparison is continued, for deep sigmoid stacks, in Exercise 3.

Solution to (e)

The map φ is the composition of L with the affine path $\eta \mapsto \theta - \eta g$. The chain rule in several variables gives

$$\varphi'(0) = \langle \nabla L(\theta),\, -g \rangle = -\langle g, g \rangle = -\|g\|_2^2.$$

If $g \neq 0$, then $\varphi'(0) < 0$. By the definition of the derivative there exists $\eta_0 > 0$ such that

$$\frac{\varphi(\eta) - \varphi(0)}{\eta} < 0 \qquad \text{for all } \eta \in (0, \eta_0),$$

hence $\varphi(\eta) < \varphi(0)$. Differentiability of L, which holds here because σ is C^∞, is the only regularity used. No global smoothness constant is required for a sufficiently small step.

On the numerical example the parameter gradient is not zero: already $\partial L/\partial b^{(2)} = -0.110824$. Its squared Euclidean norm, summing the squares of all nine scalar partials computed in (b) and (c), is

$$\|g\|_2^2 = 0.021062.$$

The step $\eta = 1/2$ therefore has first-order predicted decrease $\eta\|g\|_2^2 = 0.010531$. The updated parameters are

$$W^{(1)} \leftarrow \begin{pmatrix} 0.505071 & -0.200000 \\ 0.093073 & 0.300000 \end{pmatrix}, \qquad b^{(1)} \leftarrow \begin{pmatrix} 0.105071 \\ -0.106927 \end{pmatrix},$$

$$W^{(2)} \leftarrow \begin{pmatrix} 0.435777 & -0.472294 \end{pmatrix}, \qquad b^{(2)} \leftarrow 0.255412.$$

The second column of $W^{(1)}$ is unchanged, as its gradient vanished. The new forward pass on the same pair is

$$z^{(1)} = \begin{pmatrix} 0.610142 \\ -0.013853 \end{pmatrix}, \qquad a^{(1)} = \begin{pmatrix} 0.647973 \\ 0.496537 \end{pmatrix},$$

$$z^{(2)} = 0.303273, \qquad \hat{y} = 0.575242, \qquad L = 0.090210.$$

The loss falls from 0.100407 to 0.090210, a decrease of 0.010197. The gap between this value and the predicted 0.010531 is the usual second-order remainder: the first-order calculation uses the gradient at the old point, and L is not linear. The sign agrees, and $\eta = 1/2$ is inside the descent range for this particular loss.

References

[1] D. E. Rumelhart, G. E. Hinton, and R. J. Williams. Learning representations by back-propagating errors. *Nature*, 323:533–536, 1986.

[2] I. Goodfellow, Y. Bengio, and A. Courville. *Deep Learning*. MIT Press, 2016. Chapter 6 treats the same back-propagation identities in the column-vector convention used here.